

Vertex, Axis, Intercepts, and Opening Direction

Opening direction, axis of symmetry, vertex, and intercepts of a parabola

Directions

- For $f(x) = ax^2 + bx + c$: the parabola opens **upward** when $a > 0$ and **downward** when $a < 0$.
 - The axis of symmetry is the vertical line $x = -\frac{b}{2a}$; the vertex sits on it, so its second coordinate is the value of f at that input.
 - The y -intercept is $f(0)$. The x -intercepts are the inputs that make $f(x) = 0$ — factor and use the zero-product property.
 - Write coordinates as ordered pairs and lines as equations, e.g. $x = 3$.
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1. Let $f(x) = x^2 - 6x + 5$.

(a) Find the axis of symmetry. $x = \underline{\hspace{2cm}}$

(b) Find the second coordinate of the vertex. $y = \underline{\hspace{2cm}}$

2. Let $g(x) = x^2 - 2x - 8$.

(a) Find the x -intercepts. $x = \underline{\hspace{2cm}}$

(b) Find the y -intercept. $y = \underline{\hspace{2cm}}$

3. Let $h(x) = -3x^2 + 12x - 5$.

(a) Does this parabola open upward or downward? Justify your answer from the rule.

(b) Find the axis of symmetry. $x = \underline{\hspace{2cm}}$

4. Let $f(x) = 2x^2 + 12x + 13$.

(a) Find the axis of symmetry. $x = \underline{\hspace{2cm}}$

(b) Find the second coordinate of the vertex. $y = \underline{\hspace{2cm}}$

5. Let $f(x) = x^2 + 4x - 12$.

(a) Find the x -intercepts. $x = \underline{\hspace{2cm}}$

(b) Find the y -intercept. $y = \underline{\hspace{2cm}}$

6. Let $f(x) = (x - 4)^2 - 9$. This form shows the vertex without any rearranging.

(a) Find the second coordinate of the vertex. $y =$ _____

(b) Find the y -intercept. $y =$ _____

7. Let $f(x) = -x^2 + 6x - 8$.

(a) Find the x -intercepts. $x =$ _____

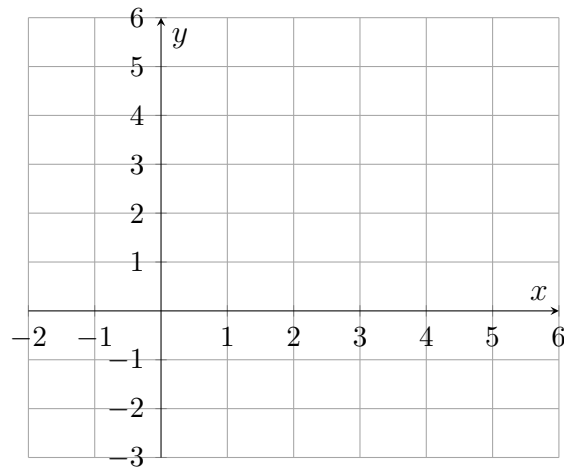
(b) Find the y -intercept. $y =$ _____

8. Let $f(x) = x^2 - 4x + 3$.

(a) Axis of symmetry: $x =$ _____ (b) Vertex y -value: _____

(c) x -intercepts: _____

(d) Sketch the parabola on the grid, and label the vertex, the axis of symmetry, and both x -intercepts.



9. Let $f(x) = 3x^2 - 18x + 20$.

(a) Find the axis of symmetry. $x =$ _____

(b) Find the second coordinate of the vertex. $y =$ _____

(c) Is that vertex a maximum or a minimum? Explain how you can tell without graphing.

10. Let $f(x) = -2x^2 + 8x + 10$.

(a) Does the parabola open upward or downward, and is its vertex a maximum or a minimum? Justify both answers.

(b) Axis of symmetry: $x =$ _____ (c) Vertex y -value: _____

(d) x -intercepts: _____